

An RNA language-model structure predictor anticipates the biochemically established P1/P1.1 mispairing of the CPEB3 human/chimpanzee ribozyme pair

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Abstract

The mammalian CPEB3 ribozyme is one of very few sequence pairs in the recently released Stanford RNA 3D Folding benchmark that differ by a single nucleotide (human R1107 has A at position 30, chimpanzee R1108 has G) yet display a ~ 4 -fold difference in self-cleavage activity *in vitro*. Skilandat *et al.* (*RNA*, 2016) attributed this gap to a P1/P1.1 mispairing in the human variant. We asked whether a state-of-the-art RNA 3D structure predictor — DRfold2 (Lee *et al.*, *PLOS Biology*, 2025), whose front end is a transformer-based RNA Composite Language Model — recovers this anchor without functional or kinetic supervision. Comparing predicted structures, the largest per-residue deviation between the human and chimpanzee predictions sits at the P1 helix endpoints (residues 9 and 60), exactly the region implicated by the biochemical model. Across five arbitrary single-nucleotide control mutations in the same sequence, this P1 anchor never appears in the top-5 most-divergent residues (**0/5**), establishing the specificity of the signal. The same comparison run with RhoFold+ in single-sequence mode shows the opposite pattern — a fixed structural prior that ignores the input mutation entirely — indicating that the “anchor recovery” property is specific to language-model-based RNA predictors rather than a property of transformer architectures in general.

1 Introduction

The structure of an RNA molecule is set by its sequence, but the *mechanism* by which a single-nucleotide change tunes a ribozyme’s activity is rarely predictable from sequence alone. The CPEB3 ribozyme — a self-cleaving HDV-like ribozyme embedded in the second intron of the human CPEB3 gene and implicated in mammalian object-location memory [5] — provides a rare, clean test. Its human and chimpanzee variants differ at exactly one position (R1107 A30, R1108 G30), and Skilandat, Boese & Sigel [3] showed biochemically that the human sequence cleaves $\sim 4\times$ slower because of a P1/P1.1 helix mispairing.

The Stanford RNA 3D Folding Kaggle benchmark (closed 2025-09-24) released ground-truth structures and a CASP-15-derived test set. The two ribozyme entries in its validation and test splits are precisely R1107 and R1108. That makes the benchmark, by accident, a clean discrimination probe: any ribozyme metric reported on its validation/test set is *de facto* a metric on the CPEB3 pair.

Recent benchmarks report substantial cross-method variability on this target — DRfold2 [1] achieves RMSD ≈ 2.72 Å on the chimpanzee crystal (PDB 7QR3), while AlphaFold3 reports RMSD ≈ 7.98 Å on the same target. What has not been examined, to our knowledge, is whether the *predicted structural response* to the 1-nt difference recovers the biochemical mechanism, not just the static fold.

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This work asks: **does a single-sequence RNA structure predictor place its largest predicted structural difference between R1107 and R1108 at the P1 helix endpoints — the residues that biochemistry says matter?**

2 Materials and methods

2.1 Targets and reference

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R1107 (human, 69 nt):
    GGGGGCCACAGCAGAAGCGUUCACGUCGCGGCCCCUGUCAGCCAUUGCACUCCGGCUGCGAAUUCUGCU
R1108 (chimp, 69 nt):
    GGGGGCCACAGCAGAAGCGUUCACGUCGCGGCCCCUGUCAGCCAUUGCACUCCGGCUGCGAAUUCUGCU
                                     ^pos 30: A->G is the
                                     only difference
```

Reference structure: PDB **7QR3** (chimpanzee CPEB3 X-ray, 2.18 Å, Przytula-Mally *et al.* [4]), chain C.

2.2 Predictors

- **DRfold2** [1]: ensemble of four configuration models (cfg_95/96/97/99); for each input sequence the pipeline produces ≈ 17 candidate structures per config, then selects and geometrically optimizes a best model. The encoder is the *RNA Composite Language Model* (RCLM) — a transformer trained on unlabeled RNA sequences. No multiple sequence alignment (MSA) is required.
- **RhoFold+** [2] run in single-sequence mode (`-single_seq_pred True`), so the architecture defaults to its sequence-only branch without MSA input.

Both predictors run on an NVIDIA RTX 5090 (32 GB VRAM) via SSH on a RunPod container. DRfold2 inference takes ≈ 3 min per 69-nt sequence; RhoFold+ single-sequence inference takes ≈ 1 s.

2.3 Null controls

We construct five single-nucleotide variants of R1108 by changing one base at each of the following peripheral positions:

ID	Mutation	Location
null_pos5	pos 5 G→A	5'-end, P1 lead
null_pos20	pos 20 U→A	P2 mid
null_pos41	pos 41 A→C	J3-P4 (peripheral)
null_pos55	pos 55 U→A	P1 3'-stem
null_pos64	pos 64 U→A	P1 3'-end

Each control receives identical DRfold2 inference as R1107/R1108.

2.4 Comparison metric

For each pair (variant prediction, R1108 prediction) we apply Kabsch alignment on C1' atoms and compute the per-residue deviation. We then extract the **top-5 most-divergent residues** for each comparison. A target is said to *hit the P1 anchor* if residue 9 *or* residue 60 (the two endpoints of the P1 helix identified by Skilandat *et al.* 2016) appears in its top-5.

2.5 Secondary structure

We compute MFE secondary structures with **ViennaRNA 2.x** [6] and also extract the RhoFold+ predicted secondary structure (**ss.ct** output) for both human and chimpanzee variants.

2.6 Reproducibility

All code, the analysis pipeline shell script, and the analysis Python heredocs are committed at <https://github.com/Vince-Anduril/rna3d-calibration>. The full run including all five null controls fits within ≈ 30 min of RTX 5090 time ($\sim \text{€}0.30$ at quoted spot pricing).

3 Results

3.1 DRfold2 places the maximal predicted divergence at the P1 helix endpoints

The Kabsch-aligned C1' RMSD between the DRfold2 predictions for R1107 and R1108 is $1.858 \text{ \AA}$ — modest but well above the model’s typical convergence noise ($\approx 0.3 \text{ \AA}$ on identical inputs). The per-residue deviation peaks not at the mutation site (position 30, $\Delta C1' = 2.15 \text{ \AA}$) but at residues forming the **P1 helix endpoints**: position 9 ($\Delta C1' = 4.21 \text{ \AA}$) and position 60 ($\Delta C1' = 4.12 \text{ \AA}$), together with three residues in the central core (positions 22, 24, 51). The top-5 most-divergent residues are therefore {9, 22, 24, 51, 60}.

3.2 The P1 anchor pattern is specific (multi-null control)

We replicated the same comparison after substituting five arbitrary single nucleotides at peripheral positions (Table 1). None of the five controls reproduces the P1 anchor: in 0/5 cases does residue 9 or 60 appear in the top-5 of the predicted divergence. In every case the top-5 is concentrated *around the local position of the mutation*. The fact that the only mutation that DOES hit the P1 endpoints is precisely the one with a documented biochemical mechanism at that exact site is the material observation.

Table 1: Multi-null robustness statistic (DRfold2 predictions, Kabsch-aligned C1' deviations vs the R1108 prediction).

Target	Pos.	mean Δ	max Δ	$\Delta@9$	$\Delta@60$	Top-5	P1?
R1107	30	1.64	4.21	4.21	4.12	{9,22,24,51,60}	YES
null_pos5	5	2.22	5.79	1.57	4.14	{1,2,3,22,23}	no
null_pos20	20	1.30	3.85	0.78	1.90	{23,24,26,48,50}	no
null_pos41	41	1.69	5.47	1.48	2.23	{48,49,50,51,52}	no
null_pos55	55	4.02	10.39	2.03	3.72	{22,23,24,25,51}	no
null_pos64	64	2.40	8.95	1.85	1.76	{50,64,65,66,67}	no

3.3 Secondary-structure-level evidence

ViennaRNA’s MFE folding gives *different* dot-bracket secondary structures for R1107 and R1108 (40 residues with different base-pairing state between the two), and the MFE of the human variant is $+1.00 \text{ kcal/mol}$ higher (less stable) than the chimpanzee. The direction of the stability gap matches the biochemical expectation that the human variant supports a weaker P1.1 (and hence less productive catalysis).

RhoFold+’s predicted secondary structure (**ss.ct** file) for the two variants also differs at ≥ 6 base pairs. In particular, RhoFold+’s human prediction contains a pair (5, 7) that is absent in the chimpanzee prediction.

3.4 The single-sequence RhoFold+ behaves qualitatively differently

In single-sequence mode, RhoFold+ reaches $\text{RMSD} = 7.7 \text{ \AA}$ vs the chimpanzee crystal — substantially worse than DRfold2. Its per-residue deviation between R1107 and R1108 predictions is dominated by the same five residues ($\{21, 48, 49, 50, 51\}$) regardless of which mutation we apply: the top-5 divergent residues for the REAL case (pos 30) and for the NULL control (pos 41) are *identical*. This is the opposite of DRfold2’s behavior. The most parsimonious interpretation is that RhoFold+ without MSA relies primarily on a fixed structural prior that swamps the small mutation-induced signal, whereas DRfold2’s language-model encoder *does* propagate the input change in a non-uniform way.

This is a cross-predictor calibration finding: the *kind* of response a predictor produces to a small input perturbation can be more informative than its raw RMSD ranking.

4 Discussion

4.1 What the language model “saw”

DRfold2’s encoder is a transformer trained, in self-supervised fashion, on a large corpus of unlabeled RNA sequences. It was not given the CPEB3 cleavage rate, the Skilandat 2016 biochemical model, or the crystal structure of any human CPEB3 (none has been released to the best of our knowledge). The fact that it nonetheless places the structural consequence of the single-nucleotide human/chimpanzee difference at the very residues identified biochemically as the seat of the activity gap is consistent with the broader observation that self-supervised language models on biological sequences capture mechanistic structure that does not appear explicitly in their training labels.

4.2 Relevance to NLP transformer interpretability

Methodologically, this work mirrors the *probing* tradition in natural language processing, where one asks “what implicit knowledge has the model acquired during pre-training, even though it was never explicitly told?”. The P1 anchor here functions as the structural-biology equivalent of a syntactic or factual probe: the input perturbation is minimal (a single nucleotide / a single token), the prediction is high-dimensional (3D coordinates / continuous embeddings), and the question is whether the model’s *response* to the small input change aligns with what an external, mechanistically grounded analysis expects. The pattern we report (specificity to the biologically meaningful mutation, indifference to arbitrary ones) is the structural-biology analogue of “the model knows the agreement rule, not just the surface form”.

4.3 Limitations

- Only one model (DRfold2) was used to establish the anchor and one (RhoFold+) for contrast. AlphaFold3, Boltz-1, Chai-1 and trRosettaRNA2 should be added before generalizing across language-model RNA predictors.
- The null control is at single-nucleotide resolution, but only five mutations were tested; a larger, statistically powered control set (e.g., 20 random single-nucleotide variants) would tighten the specificity claim.
- We did not directly inspect DRfold2’s per-residue confidence (its `.ret` files are pickled dictionaries whose schema we have not yet exhaustively decoded); doing so would test whether the model is *confident* about the cascade residues, which is the natural next probe.
- Re-running RhoFold+ *with* a proper MSA (rather than single-sequence mode) would clarify whether the fixed-prior failure is an artifact of the no-MSA branch or a property of the architecture proper.

4.4 What this enables

Beyond CPEB3, the protocol — *pair a known single-nucleotide functional variant with predictions of a sequence-only structure model and look at the predicted-divergence cascade* — is a low-cost, model-agnostic probe of whether a structural predictor has internalized mechanistic information. It requires no new training, no MSA, no functional labels, and runs in minutes on a single GPU.

5 Data and code availability

All sequences, scripts, intermediate predictions, and analysis tables are available at <https://github.com/Vince-Anduril/rna3d-calibration>.

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References

- [1] Lee *et al.* “DRfold2: deep learning RNA structure prediction.” *PLOS Biology* (2025).
- [2] Shen *et al.* “Accurate RNA 3D structure prediction using a language model-based deep learning approach.” *Nature Methods* 21, 1545–1554 (2024).
- [3] Skilandat, Boese & Sigel “Secondary structure confirmation and localization of Mg^{2+} ions in the mammalian CPEB3 ribozyme.” *RNA* 22, 750–763 (2016).
- [4] Przytula-Mally *et al.* “Anticodon-like loop-mediated dimerization in the crystal structures of HDV-like CPEB3 ribozymes.” bioRxiv 2022.09.22.508989 (2022).
- [5] Vogler *et al.* “Inhibition of CPEB3 ribozyme elevates CPEB3 protein expression and polyadenylation of its target mRNAs, and enhances object location memory.” PMID:38319152 (2024).
- [6] Lorenz *et al.* “ViennaRNA Package 2.0.” *Algorithms for Molecular Biology* 6, 26 (2011).
- [7] Salehi-Ashtiani *et al.* “A genomewide search for ribozymes reveals an HDV-like sequence in the human CPEB3 gene.” *Science* 313, 1788–1792 (2006).
- [8] Guo, Pleiss, Sun & Weinberger. “On calibration of modern neural networks.” *ICML* (2017). arXiv:1706.04599.
- [9] Riccitelli *et al.* “Experimental resurrection of ancestral mammalian CPEB3 ribozymes reveals deep functional conservation.” *MBE* 38, 2843–2861 (2021).
- [10] Critical Assessment of Structure Prediction (CASP15) — RNA target highlights. *Proteins* (2023).